# THIAMINE

Give BEFORE glucose

Anaphylaxis with IV

**CHEMICAL NAME** 3-[(4-amino-2-methylpyrimidin-5-yl)methyl]-5-(2-hydroxyethyl)-4-methyl-1,3-thiazol-3-ium chloride (hydrochloride)

**CLASS** Water-soluble vitamin B1

**TRADE NAMES** Pabrinex (IV, with other B/C vitamins) · Benerva · Thiamine HCl

## THERAPEUTIC USES / INDICATIONS

- Wernicke's encephalopathy — treatment and prevention (alcohol misuse, malnutrition, hyperemesis gravidarum, bariatric surgery).
- Thiamine deficiency (beriberi; wet and dry).
- Refeeding syndrome prevention.
- Alcohol withdrawal (prophylaxis).

## ADMINISTRATION

- **IV** (Pabrinex ampoules 1+2 diluted in 50–100 mL saline over 30 min) or **IM**.
- Give before or with glucose in at-risk patients (glucose without thiamine can precipitate Wernicke's).
- **PO** tablets for maintenance.

## DOSAGES

- **Suspected/confirmed Wernicke's:** Pabrinex 2 pairs (≈500 mg thiamine) IV three times daily for 3–5 days, then 1 pair daily for 3–5 days.
- **Prophylaxis (alcohol misuse):** Pabrinex 1 pair IV daily for 3–5 days, or oral thiamine 100 mg three times daily.
- **Mild deficiency / maintenance:** 25–100 mg PO daily.

## PHARMACOKINETICS

- Oral absorption limited (saturable, ↓ by alcohol).
- Converted to active thiamine pyrophosphate.
- Excess excreted in urine. **t½:** ~1–2 h (body stores 2–3 weeks).

## NURSING MANAGEMENT

- Identify at-risk patients (alcohol, malnutrition, vomiting) on admission.
- Recognise Wernicke's triad: confusion, ataxia, ophthalmoplegia/nystagmus (often incomplete).
- Infuse slowly; observe for allergy.
- Give thiamine before IV glucose in at-risk patients.

## PHARMACODYNAMICS

- Thiamine pyrophosphate is a coenzyme in **carbohydrate metabolism** (pyruvate dehydrogenase, α-ketoglutarate dehydrogenase, transketolase) → essential for brain and heart energy.

## SIDE & ADVERSE EFFECTS

### SIDE EFFECTS (COMMON)

- Injection-site pain.
- Warmth, tingling.
- Nausea.

### ADVERSE EFFECTS (SERIOUS)

- Anaphylaxis (rare, IV Pabrinex); give over 30 min with resuscitation facilities.

## CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to thiamine or Pabrinex components.

### KEY INTERACTIONS

- **Alcohol:** ↓ absorption.
- **Loop diuretics:** ↑ thiamine loss (heart failure).

## PATIENT EDUCATION

- Reduce or stop alcohol; seek support.
- Take thiamine tablets daily as prescribed.
- Eat balanced meals.

## KEY POINTS

- Vitamin B1; carbohydrate metabolism.
- Thiamine before glucose in alcohol misuse.
- Wernicke's: high-dose IV Pabrinex.
- Anaphylaxis risk with IV — infuse over 30 min.

**SOURCES CHECKED** **[940]** Pabrinex/Vitamin B&C Intravenous High Potency, Concentrate for Solution for Infusion (UK SmPC) **[941]** thiamine hydrochloride, IM, IV (US label) **[942]** Thiamine Hydrochloride, IM, IV (US label)

*Full references p. 432–457*